The Silver Rate Is (Almost) Tight

The silver exponents are optimal for gradient descent with predetermined nonnegative stepsizes, in both the non-anytime and anytime settings.

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Research blog version of Silver Rate Is (Almost) Optimal for Gradient Descent (arXiv:2609.09152). Updated November 2, 2026. Download this post as a PDF.

Abstract

We study gradient descent with predetermined nonnegative stepsizes on smooth convex functions. We prove lower bounds that match the silver polynomial convergence exponent in the non-anytime setting and the best known polynomial exponent in the anytime setting, up to a subpolynomial loss. The construction bends the local gradient trajectory in a checkpoint-based hard function and combines the resulting transfer bound with a weighted splitting argument. The exact endpoint rates remain open.

We're still working to simplify the proofs and explain the intuition as clearly as possible, which is not easy for me (unfortunately). We'll update this post as the exposition improves.

Silver stepsizes accelerate gradient descent beyond the classical $O(1/n)$ rate using stepsizes alone. In this post, we show that no predetermined nonnegative stepsize schedule can improve on the silver polynomial convergence exponent. This result was developed through nontrivial interactions with ChatGPT 6 Astra Ultra (see AI Disclosure). Our arXiv preprint is now available.

Introduction

Let

$$p_{\mathrm{sil}} := \log_2(1+\sqrt{2}) = 1.271553303\ldots.$$

Altschuler and Parrilo (2023) first proved the non-anytime rate $O(n^{-p_{\mathrm{sil}}})$ for gradient descent on smooth convex functions using their silver stepsize schedule. They also conjectured that this rate is optimal up to constant factors. Building on their recursive gluing technique, Grimmer, Shu, and Wang subsequently developed compositions of stepsize schedules that achieve the same exponent with improved constants. For a broader introduction to stepsize-based acceleration, see Altschuler and Parrilo's recent survey.

Recent lower bounds have moved rapidly toward the silver exponent. Ma and Chen obtained $\Omega(n^{-1.932})$, Tsai improved the exponent to 1.7321, and in our previous blog post, we presented the $\Omega(n^{-1.6342})$ result from our earlier paper. Jung et al. then introduced a new checkpoint construction and reached $\Omega(n^{-\log_2(1+\sqrt{3})}) = \Omega(n^{-1.4500})$.

The new lower bound reaches $n^{-p_{\mathrm{sil}}-o(1)}$, closing the remaining gap in the polynomial exponent. The same construction also closes the anytime gap at the level of polynomial exponents.

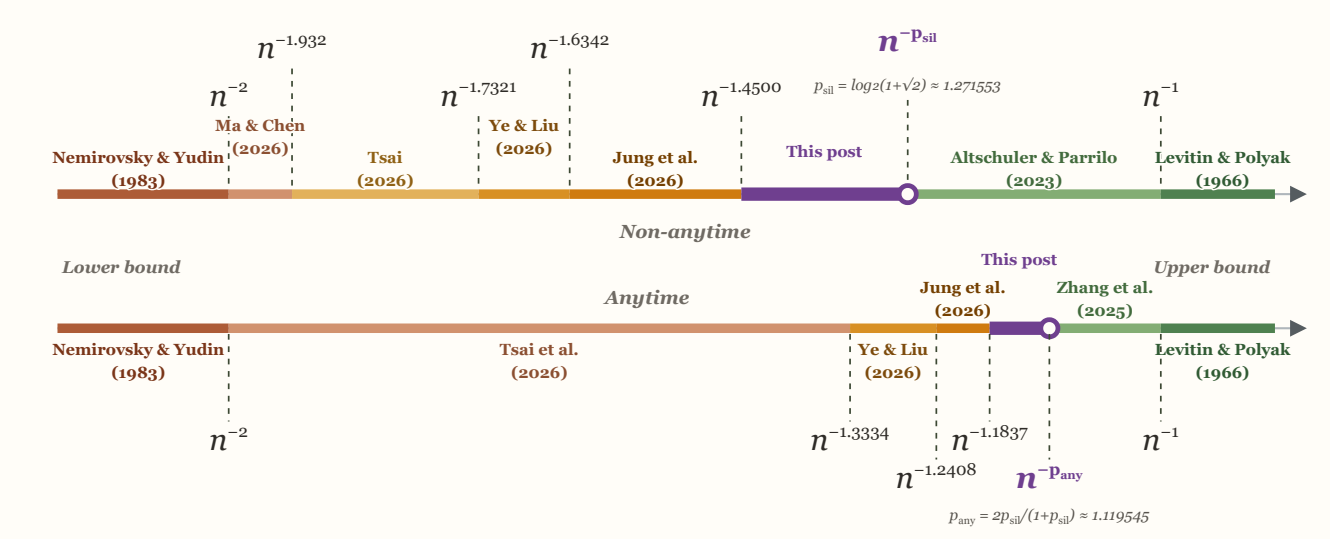


Figure 1. Lower and upper bounds on the optimal polynomial convergence exponents for non-anytime and anytime gradient descent.

Problem Setup and Main Results

For a predetermined schedule $h = (h_1, \dots, h_n) \in [0, \infty)^n$, gradient descent generates $x_t = x_{t-1} - h_t \nabla F(x_{t-1})$. After scaling the smoothness constant to one, define

$$\mathcal{R}_n(h) := \sup_{F, x_0} \frac{F(x_n) - F(x_*)}{\|x_0 - x_*\|^2},$$

where the supremum ranges over finite-dimensional convex 1-smooth functions with a minimizer x_*, and over initial points $x_0 \neq x_*$. Let $r_n^* := \inf_{h \in [0,\infty)^n} \mathcal{R}_n(h)$.

> **Theorem 1.** *There is an absolute constant $C > 0$ such that, for every sufficiently large n,*
>
> $$\boxed{r_n^* \geq n^{-\left(p_{\mathrm{sil}} + C\sqrt{\frac{\log\log n}{\log n}}\right)}.} \tag{1}$$

Together with the silver-schedule upper bound of Altschuler and Parrilo, this determines p_{sil} as the optimal polynomial convergence exponent.

For an infinite nonnegative schedule $H = (h_t)_{t \geq 1}$, write $H_n = (h_1, \dots, h_n)$, and set $p_{\mathrm{any}} := 2p_{\mathrm{sil}}/(1 + p_{\mathrm{sil}}) = 1.119545204\ldots$.

Theorem 2. *There is an absolute constant $C > 0$ such that every infinite nonnegative schedule has infinitely many horizons n for which*

$$\boxed{\mathcal{R}_n(H_n) \geq n^{-\left(p_{\mathrm{any}} + C\sqrt{\frac{\log\log n}{\log n}}\right)}.} \tag{2}$$

In particular, no anytime schedule has a rate $O(n^{-q})$ for any $q > p_{\mathrm{any}}$. This matches the polynomial exponent in the anytime upper bound of Zhang et al. Hence p_{any} is the optimal anytime polynomial convergence exponent.

A Sharper Hard-Function Bound

We choose checkpoints $1 \leq t_1 < \cdots < t_k \leq n$, set $t_0 := 0$ and $t_{k+1} := n + 1$, and write

$$b_i := h_{t_i} \quad (1 \leq i \leq k), \qquad s_i := \sum_{j=t_{i-1}+1}^{t_i-1} h_j \quad (1 \leq i \leq k+1).$$

Thus, for $i \leq k$, s_i is the sum of the steps strictly between checkpoints t_{i-1} and t_i, while s_{k+1} is the sum after the last checkpoint. Fix $\epsilon, \gamma > 0$, and set $c := \sqrt{1 + 2\gamma + \epsilon^2}$.

Theorem 3. *There are finite constants $a = a(\epsilon, \gamma)$ and $B_0 = B_0(\epsilon)$ such that, whenever every selected checkpoint satisfies $b_i \geq B_0$,*

$$\boxed{\mathcal{R}_n(h) \geq \frac{1}{4(1 + s_{k+1})} \prod_{i=1}^{k} \left[\frac{b_i}{c(a + s_i) + \epsilon b_i}\right]^2.} \tag{3}$$

For $k = 0$, the product is empty and $s_{k+1} = \sum_{t=1}^{n} h_t$.

We use a chain of local hard functions, as in Jung et al. The i-th component uses only the adjacent coordinates (x_i, x_{i+1}). It receives the remaining amplitude in x_i, and the checkpoint step $b_i = h_{t_i}$ transfers that amplitude into x_{i+1}, where the next component takes over. In their construction, the active Huber component remains affine during the gap and contributes a gradient in a fixed direction. In our construction, the projected gradient instead rotates along a circular arc, producing a bent iterate path. This lowers the coefficient paid for each gap from 2 to a value arbitrarily close to 1.

The price of reducing the gap coefficient to c, which can be made arbitrarily close to one, is the additional ϵb_i term. We cannot simply discard this term, so the global argument must account for it exactly.

More precisely, during each gap the projected gradient takes values on a circular arc with endpoints $(c, 0)$ and $(\epsilon, -1)$, reaching the latter just before the checkpoint. Choosing these directions backward from the checkpoint produces a valid gradient-descent trajectory for the actual gap steps. The steps during which the direction turns have bounded total stepsize. All remaining gap steps use the horizontal direction $(c, 0)$. A checkpoint of size b therefore transfers an incoming amplitude D to

$$D\,\frac{b}{c(a+s)+\epsilon b}.$$

A Moreau envelope realizes these prescribed gradients, and the old two-coordinate component remains inactive after the transfer. Chaining the local components gives (3). The use of a Moreau envelope to realize prescribed gradients also appears in Ma and Chen's hard-function construction.

Full proof of Theorem 3

If $k = 0$, the one-dimensional Huber function gives $\mathcal{R}_n(h) \geq 1/[4(1 + \sum_t h_t)]$, which is the stated formula. We henceforth assume $k \geq 1$.

1. A curved set of possible gradients

Set

$$R := \sqrt{\epsilon^2 + (1+\gamma)^2}, \qquad p(q) := \sqrt{R^2 - (q+\gamma)^2}, \qquad v(q) := (p(q), -q), \quad 0 \leq q \leq 1.$$

The vectors $v(q)$ form a circular arc centered at $(0, \gamma)$, with endpoints $v(0) = (c, 0)$ and $v(1) = (\epsilon, -1)$. Along the arc, $\epsilon \leq p(q) \leq c$ and $q/p(q) \leq 1/\epsilon$. Let

$$K := \operatorname{conv}\big(\{0\} \cup \{v(q) : 0 \leq q \leq 1\}\big).$$

If σ_K is the support function of K, then its Moreau envelope satisfies $\nabla \operatorname{env}_1 \sigma_K(r) = \Pi_K(r)$. Indeed, the projection condition $\langle r - v, w - v\rangle \leq 0$ for all $w \in K$ is exactly the optimality condition $v \in \partial \sigma_K(r - v)$. Since projection is nonexpansive, the envelope is convex and 1-smooth.

2. Choosing the gradients backward

Consider one gap with nonnegative steps $h_1, \ldots, h_m$ and total size $S := \sum_{j=1}^m h_j$. Start backward from $r_m = (\epsilon, -1)$, with projected gradient $v_m = (\epsilon, -1)$. Suppose $r_j = (u, -y)$, put $h = h_j$, and define $\Delta := y + \gamma - h\gamma$. If $\Delta \leq u\gamma/c$, choose $v_{j-1} = (c, 0)$. Otherwise set

$$D := \sqrt{u^2 + \Delta^2}, \qquad p := \frac{Ru}{D}, \qquad q := \frac{R\Delta}{D} - \gamma, \qquad v_{j-1} := (p, -q).$$

In either case, let $r_{j-1} := r_j + h_j v_{j-1}$. We claim that

$$v_j = \Pi_K(r_j), \qquad 0 \leq j \leq m. \tag{4}$$

The recursion preserves $u \ge \epsilon$, $y \ge 1$, and $y/u \le 1/\epsilon$. Indeed, after a backward step, the new ratio is a weighted average of y/u and q/p, both at most $1/\epsilon$. In the second case, these inequalities imply $0 < q \le 1$, and direct substitution gives

$$r_{j-1} - v_{j-1} = \left(\frac{D}{R} + h - 1\right)\big(v_{j-1} - (0,\gamma)\big).$$

The coefficient is nonnegative. This is immediate for $h \ge 1$, while for $h \le 1$, $D \ge R - h\gamma$. The second factor is the outward normal to the circle and supports all of K, so the projection condition holds. When the chosen direction is horizontal, writing $\mu := u/c + h - 1$, the condition for this case gives $0 \le y \le \gamma\mu$, and $r_{j-1} - (c,0) = (c\mu, -y)$ lies in the normal cone of K at $(c,0)$. Once a horizontal direction is chosen, all earlier gradients can remain $(c,0)$. This proves (4).

3. Bounding the total stepsize spent turning

During a curved backward step, let $\sigma := (y+\gamma)/u$. If σ_h and σ_ℓ are its values before and after that step, then

$$\sigma_\ell = \sigma_h - \frac{h\gamma}{u}, \qquad \frac{u_{\text{new}}}{u_{\text{old}}} = 1 + \frac{R}{\gamma}\frac{\sigma_h - \sigma_\ell}{\sqrt{1+\sigma_\ell^2}}.$$

For $A(\sigma) := \sigma + \sqrt{1+\sigma^2}$, convexity and Bernoulli's inequality imply

$$\frac{u_{\text{new}}}{u_{\text{old}}} \le \left[\frac{A(\sigma_h)}{A(\sigma_\ell)}\right]^{R/\gamma}.$$

These ratios telescope. Initially $\sigma = (1+\gamma)/\epsilon$, and at the other end of the curved part $\sigma > \gamma/c$. Thus $u_{\text{end}} \le \epsilon[A((1+\gamma)/\epsilon)/A(\gamma/c)]^{R/\gamma}$. On the other hand, every curved gradient has first component at least ϵ, so $u_{\text{end}} \ge \epsilon(1 + H_{\text{turn}})$. Hence

$$H_{\text{turn}} \le H := \left[\frac{c(1+\gamma+R)}{\epsilon(\gamma+R)}\right]^{R/\gamma} - 1. \tag{5}$$

Write $P := \sum_j h_j p_{j-1}$ and $Q := \sum_j h_j q_{j-1}$. Horizontal steps do not contribute to Q, and curved steps contribute at most their stepsize. Therefore $P \le cS$, $Q \le H$, and $r_0 = (\epsilon + P, -1 - Q)$. Define

$$C_0 := \epsilon + \epsilon^{-1} + H/\epsilon,$$

so that $\epsilon + P + (1+Q)/\epsilon \le cS + C_0$.

4. Scaling the local function

Let the local checkpoint have size $b \ge 1 + \epsilon^{-2}$, choose any offset $C_{\text{off}} \ge C_0$, and set

$$\rho := \frac{1}{cS + \epsilon b + C_{\mathrm{off}}}, \qquad z_\circ := (1,0) - \rho r_0.$$

Let $\mathcal{V}$ be the finite set of gradients used above, including $(c,0)$, and define

$$\phi(z) := \max\{0, \rho\langle v, z - z_\circ\rangle : v \in \mathcal{V}\}, \qquad f := \mathrm{env}_1\, \phi.$$

The projection conditions remain valid after the unused arc directions are removed, so $\nabla f(z_\circ + \rho r_j) = \rho v_j$. The bound above also shows that every affine piece of ϕ is nonpositive on $\{z_1 \le 0, z_2 \ge 0\}$, and hence $f = 0$ there.

The normalized gap starts at $(1,0)$. Just before the checkpoint its gradient is $\rho(\epsilon, -1)$. Just after the checkpoint its position is

$$(U, Y + \rho b), \qquad U := 1 - \rho(P + \epsilon b) \ge 0, \qquad Y := \rho Q.$$

For every used direction $v = (p, -q)$, $\langle v, (U,Y) - z_\circ\rangle = \rho((1-b)\epsilon p + q) \le 0$, where we used $q/p \le 1/\epsilon$ and $b \ge 1 + \epsilon^{-2}$. Increasing the second coordinate only decreases this inner product. Thus the old component is zero forever after the checkpoint, while the output passed to the next component has size

$$G(S,b) = \rho b = \frac{b}{cS + \epsilon b + C_{\mathrm{off}}}. \tag{6}$$

5. Chaining the local components

Take

$$B_0 := 2(1 + \epsilon^{-2}), \qquad a \ge \max\{2, 2C_0/c, 2B_0\}.$$

For the i-th gap, apply the local construction to steps $h_t/2$, checkpoint $b_i/2$, and offset $C_{\mathrm{off}} = ca/2$. This is admissible by the preceding choice of a, B_0, and (6) becomes exactly

$$G_i = \frac{b_i}{c(s_i + a) + \epsilon b_i}.$$

Let f_i be the local two-coordinate function and Y_i its normalized activation level. Set

$$D_1 := 1, \quad \ell_1 := 0, \qquad D_{i+1} := D_i G_i, \quad \ell_{i+1} := D_i Y_i,$$

and define the scaled component

$$\Phi_i(x) := D_i^2 f_i\left(\frac{x_i - \ell_i}{D_i}, \frac{x_{i+1}}{D_i}\right).$$

For $\delta := D_{k+1}/(1 + s_{k+1})$, let

$$H_\delta(z) := \begin{cases} 0, & z \le 0, \\ z^2/2, & 0 \le z \le \delta, \\ \delta z - \delta^2/2, & z \ge \delta, \end{cases}$$

and take

$$F(x) := \frac{1}{2}\sum_{i=1}^{k} \Phi_i(x) + \frac{1}{2} H_\delta(x_{k+1} - \ell_{k+1}).$$

Earlier components stay inactive, so only the current component is active along the prescribed trajectory. Every coordinate belongs to at most two 1-smooth components, so the factor $1/2$ makes F convex and 1-smooth. All terms are nonnegative and vanish at the origin, hence $F(0) = \min F = 0$.

Starting from $x_0 = e_1$, induction over the components shows that the checkpoint output of block i is precisely the input of block $i+1$. Earlier blocks remain zero. The final Huber term stays in its affine region through the last gap and leaves value $D_{k+1}^2/[4(1+s_{k+1})]$. Since $D_{k+1} = \prod_i G_i$ and $\|x_0 - x_*\| = 1$, this is exactly (3).

□

A Cost Function for Checkpoint Selection

We adapt the checkpoint-cost framework and tight-set analysis of Jung, Cho, and Yun to the modified factors below.

Fix the constants in Theorem 3 with $0 < \epsilon < 1$, take $\lambda \ge a$, and write $d := \epsilon/c$ and $\chi := c - \epsilon = c(1-d)$. For a checkpoint set T, define

$$\Psi_\lambda(T;h) := (\lambda + s_{k+1}) \prod_{i=1}^{k} \left(\epsilon + \frac{c(a+s_i)}{b_i}\right), \quad V_\lambda(h) := \min_T \Psi_\lambda(T;h), \quad U_n(\lambda) := \sup_{h \in [0}$$

A set selecting a zero step as a checkpoint is assigned infinite cost.

Even as $b_i \to \infty$, its factor approaches ϵ, rather than zero. We need to account for this positive limit in the recursive argument. The next lemma connects the cost directly to the optimization error.

Lemma 4. *For every $h \in [0,\infty)^n$ and $\lambda \ge a$,*

$$\boxed{\mathcal{R}_n(h) \ge \frac{\lambda - 1}{V_\lambda(h)^2}.} \tag{8}$$

Full proof

Suppose a step b is added to a checkpoint set. Let ℓ be the sum of the unselected steps between the preceding checkpoint and b, or before b if it is the first checkpoint. Let r be the sum of the unselected steps strictly between b and the next checkpoint, or after b if there is no later checkpoint. Set $y = a + r + db'$ if the next selected checkpoint is b', and $y = \lambda + r$ if there is no later checkpoint. Adding b multiplies the cost by

$$I(\ell, y; b) = \frac{c(a + \ell + db)y}{b(\ell + b + y)}.$$

Both $\partial_\ell \log I$ and $\partial_y \log I$ are positive. Hence, if $b \le a/2$, $I(\ell, y; b) \ge I(0, a; b) \ge 4/3$. Deleting such a checkpoint strictly lowers the cost. An optimal set therefore uses only $b_i > a/2 \ge B_0$, so Theorem 3 applies.

For an optimal T, Theorem 3 and the definition of Ψ_λ give

$$\mathcal{R}_n(h) \ge \frac{(\lambda + s_{k+1})^2}{4(1 + s_{k+1})V_\lambda(h)^2}.$$

Finally, $(\lambda + s)^2 - 4(\lambda - 1)(1 + s) = (s - \lambda + 2)^2 \ge 0$, which proves the lemma.

□

Tight checkpoint sets

Call T *tight* if $\Psi_\lambda(T; h) = V_\lambda(h)$. The insertion factor above is increasing in both adjacent gaps. Selecting more checkpoints shortens these gaps, so adding another checkpoint multiplies the cost by a factor no larger than before. This gives the following log-submodularity inequality.

> **Lemma 5.** *For every positive schedule,*
>
> $$\Psi_\lambda(A; h)\Psi_\lambda(B; h) \ge \Psi_\lambda(A \cap B; h)\Psi_\lambda(A \cup B; h). \tag{9}$$
>
> *The inequality is strict when A and B are disjoint and nonempty.*

Thus tight sets are closed under unions and intersections, and two nonempty tight sets cannot be disjoint. A compactness argument then shows that a maximizing schedule exists.

> **Lemma 6.** *For every $n \ge 1$ and $\lambda \ge a$, $U_n(\lambda)$ is finite, strictly exceeds $U_{n-1}(\lambda)$, and is attained by a positive finite schedule. At every maximizing schedule, both the empty set and at least one nonempty checkpoint set are tight.*

Full proof of Lemmas 5 and 6

For Lemma 5, add the elements of $B \setminus A$, one at a time, to each of the sets $A \cap B$ and A. At each insertion, the adjacent gaps for the second set are no larger, so the factor multiplying its cost is no larger. Multiplication gives (9). If A, B are nonempty and disjoint, at least one adjacent gap shrinks strictly.

For Lemma 6, induct on n, starting from $U_0(\lambda) = \lambda$. If $M := \max_t h_t \le a$, the empty set gives $V_\lambda(h) \le \lambda + na$. If $M > a$, selecting the first position attaining M and optimizing its suffix gives $V_\lambda(h) \le (\epsilon + cn)U_{n-1}(\lambda)$. Thus $U_n(\lambda) < \infty$.

Appending a sufficiently small positive step to a maximizing $(n-1)$-step schedule strictly raises every cost that omits it, while every cost selecting it diverges as that step tends to zero. Hence $U_n > U_{n-1}$. A maximizing sequence cannot have a coordinate tend to infinity, since selecting the first such coordinate bounds the limsup of its cost by $\epsilon U_{n-1} < U_n$. Nor can a coordinate tend to zero, because deleting it leaves at most $n-1$ steps. The maximizing sequence therefore lies in a compact subset of $(0, \infty)^n$, and continuity gives an optimizer.

If the empty set were not tight, all tight sets would be nonempty and, by Lemma 5, would share a checkpoint. Slightly decreasing that shared step raises every tight cost while preserving the slack of all other sets, contradicting maximality. If the empty set were the only tight set, slightly increasing any step gives the same contradiction. Therefore both an empty and a nonempty tight set exist.

□

Moving Part of Each Checkpoint into Its Gap

Each checkpoint factor in (7) contains an additive ϵ. We remove this term by moving part of each selected step into the preceding gap, while preserving the cost of every checkpoint subset.

We first define an auxiliary cost. Fix gaps $g_1, \dots, g_k \ge 0$, checkpoint sizes $B_1, \dots, B_k \ge 0$, and $\Lambda \ge a$. For a nonempty set $S = \{i_1 < \cdots < i_m\} \subseteq [k]$, with $i_0 := 0$, let

$$L_j(S) := \sum_{r=i_{j-1}+1}^{i_j} g_r + \sum_{r=i_{j-1}+1}^{i_j - 1} B_r, \qquad Y(S) := \Lambda + \sum_{r > i_m} (g_r + B_r).$$

Define

$$\Phi_\Lambda(S; g, B) := Y(S) \prod_{j=1}^{m} \frac{\chi(a + L_j(S))}{B_{i_j}}, \qquad \Phi_\Lambda(\varnothing; g, B) := \Lambda + \sum_{i=1}^{k} (g_i + B_i). \tag{10}$$

If $B_i = 0$, a subset containing i is assigned infinite cost.

Now take a maximizing schedule from Lemma 6 and any nonempty tight set T, with checkpoint sizes b_i and gaps s_i. We move the fraction d of every selected step into the gap immediately before it by setting

$$\boxed{B_i := (1-d)b_i, \qquad g_i := s_i + db_i, \qquad \Lambda := \lambda + s_{k+1}, \qquad \chi := c - \epsilon.} \tag{11}$$

Proposition 7. *For every $S \subseteq [k]$,*

$$\boxed{\Phi_\Lambda(S; g, B) = \Psi_\lambda(\{t_i : i \in S\}; h).} \tag{12}$$

Full proof

If checkpoint i is skipped, its contribution to the next gap is unchanged because $g_i + B_i = s_i + b_i$. Suppose instead that i_j is selected after i_{j-1}, and let $\tilde{s}_j$ be the corresponding gap in the original schedule. The transformed gap is $L_j(S) = \tilde{s}_j + db_{i_j}$. Hence

$$\frac{\chi(a + L_j(S))}{B_{i_j}} = \frac{c(1-d)(a + \tilde{s}_j + db_{i_j})}{(1-d)b_{i_j}} = \frac{c(a+\tilde{s}_j)}{b_{i_j}} + \epsilon.$$

The final factor is also unchanged, so the two costs are equal.

□

Because the empty set and T are both tight, the empty set minimizes Φ_Λ. We also obtain identities for the smaller problems inside the gaps. If $h^{(i)}$ is the schedule in gap i and S is any checkpoint set inside that gap, direct factorization gives

$$\frac{\Psi_\lambda(T \cup S; h)}{\Psi_\lambda(T; h)} = \frac{\Psi_{a+db_i}(S; h^{(i)})}{a + s_i + db_i}.$$

The left side is at least one, while the empty set in the gap attains equality. For the final gap, the corresponding factorization is

$$\frac{\Psi_\lambda(T \cup S; h)}{\Psi_\lambda(T; h)} = \frac{\Psi_\lambda(S; h^{(k+1)})}{\lambda + s_{k+1}}.$$

The same reasoning now gives

$$\boxed{V_{a+db_i}(h^{(i)}) = a + s_i + db_i, \qquad V_\lambda(h^{(k+1)}) = \lambda + s_{k+1}.} \tag{13}$$

The Weighted Splitting Lemma

The following lemma adapts the singleton-splitting argument of Jung et al. by adding a weighted term.

We now prove a recursive bound for Φ_Λ. The extra term involving the checkpoint sizes will cancel the contribution of db_i from the parameter $a + db_i$ in (13).

Lemma 8. *Fix $\nu \in [1/2, 1)$ and $\kappa \geq 0$. Suppose that for every $x, y \geq a$ and $B > 0$, the relations*

$$w = x + y + B - a, \qquad wB = \chi xy,$$

imply

$$\boxed{w^\nu + \kappa B^\nu \leq x^\nu + y^\nu.} \tag{14}$$

If the empty set minimizes Φ_Λ and $W := \Phi_\Lambda(\varnothing; g, B)$, then

$$\boxed{W^\nu + \kappa \sum_{i=1}^{k} B_i^\nu \leq \Lambda^\nu + \sum_{i=1}^{k} (a + g_i)^\nu.} \tag{15}$$

A tight singleton gives exactly the two relations preceding (14). The main step is to show that a maximizing choice of checkpoint sizes has a tight singleton.

Full proof

We proceed by induction on k. The case $k = 0$ is immediate. Keep the gaps fixed and maximize

$$J(B) := W^\nu + \kappa \sum_{i=1}^{k} B_i^\nu$$

over all $B_i \geq 0$ for which the empty set is optimal.

The maximum is attained

The singleton constraint for checkpoint i has the form $WB_i \leq \chi x_i y_i$, where $W \geq y_i$ and x_i depends only on earlier variables. Therefore $B_i \leq \chi x_i$, which successively bounds all checkpoint sizes. After multiplying each subset constraint by the product of its selected B_i's, the feasible set is described by continuous polynomial inequalities. It is closed and bounded, so J attains a maximum.

Zero checkpoint sizes

If some $B_i = 0$, delete it and merge the adjacent gaps. The induction hypothesis applies, using $(a + g_i + g_{i+1})^\nu \le (a + g_i)^\nu + (a + g_{i+1})^\nu$. For the last checkpoint, absorb its gap into Λ and use $(\Lambda + g_k)^\nu \le \Lambda^\nu + (a + g_k)^\nu$. It remains to treat an interior maximizer with every $B_i > 0$.

A minimal tight set is a singleton

At least one nonempty subset is tight. Otherwise, slightly increasing any B_i raises J while preserving all constraints. The cost Φ_Λ has the same strict log-submodularity property as Ψ, since its insertion ratio $\chi(a + \ell)y/[B(\ell + B + y)]$ is strictly increasing in ℓ, y. Hence an inclusion-minimal nonempty tight set T_0 lies inside every nonempty tight set.

Suppose T_0 contains two unequal sizes $u < v$, and put $r = v/u > 1$. Define

$$A_* := \frac{v(W + u)}{u(W + v)}, \qquad D_* := \frac{W^{\nu-1} + \kappa u^{\nu-1}}{W^{\nu-1} + \kappa v^{\nu-1}}.$$

Since $W > u + v$ and $\nu \ge 1/2$,

$$A_* > \frac{r(r + 2)}{2r + 1} \ge \sqrt{r} \ge r^{1-\nu} \ge D_* \ge 1.$$

Choose $\theta \in (D_*, A_*)$ and perturb $u \mapsto u - t, v \mapsto v + \theta t$. For every nonempty tight set, the derivative of its cost minus the empty-set cost is $W(1/u - \theta/v) - (\theta - 1) > 0$, because $\theta < A_*$. The derivative of J is

$$\nu\big[(\theta - 1)W^{\nu-1} + \kappa(\theta v^{\nu-1} - u^{\nu-1})\big] > 0,$$

because $\theta > D_*$. All other constraints remain strict for sufficiently small t, contradicting maximality.

If the two sizes are equal to b, perturb them to $b - t$ and $b + t + t^2/(2b)$. Their product decreases by $t^2/2 + O(t^3)$, whereas W grows only by $t^2/(2b)$. Every nonempty tight constraint gains slack with leading term $(W - b)t^2/(2b^2) > 0$, while J increases by

$$\frac{\nu t^2}{2b}\big[W^{\nu-1} + \kappa(2\nu - 1)b^{\nu-1}\big] + O(t^3) > 0.$$

This is again a contradiction. Therefore $T_0 = \{r\}$ for some r.

Split at the tight singleton

Let

$$x := a + \sum_{i \le r} g_i + \sum_{i < r} B_i, \qquad y := \Lambda + \sum_{i > r} (g_i + B_i).$$

Tightness gives $W = x + y + B_r - a$ and $WB_r = \chi xy$. Define the costs on either side of r by

$$\Phi_L(S_L) := \Phi_{a+g_r}(S_L; g_{1:r-1}, B_{1:r-1}), \qquad \Phi_R(S_R) := \Phi_\Lambda(S_R; g_{r+1:k}, B_{r+1:k}),$$

where the variables to the right of r are reindexed starting from one. For $S_L \subseteq \{1, \dots, r-1\}$ and $S_R \subseteq \{r+1, \dots, k\}$, we have

$$\Phi_\Lambda(S_L \cup \{r\} \cup S_R; g, B) = \frac{\chi}{B_r} \Phi_L(S_L) \Phi_R(S_R).$$

In particular,

$$\Phi_\Lambda(S_L \cup \{r\}; g, B) = \frac{\chi y}{B_r}\, \Phi_L(S_L), \qquad \Phi_\Lambda(\{r\} \cup S_R; g, B) = \frac{\chi x}{B_r}\, \Phi_R(S_R).$$

Since the empty set minimizes Φ_Λ and $\{r\}$ is tight, the empty set is also optimal in both smaller problems. The induction hypothesis gives

$$x^\nu + \kappa \sum_{i<r} B_i^\nu \le \sum_{i \le r} (a + g_i)^\nu,$$

$$y^\nu + \kappa \sum_{i>r} B_i^\nu \le \Lambda^\nu + \sum_{i>r} (a + g_i)^\nu.$$

Adding these inequalities to the scalar hypothesis $W^\nu + \kappa B_r^\nu \le x^\nu + y^\nu$ proves (15).

□

Exact Cancellation

We choose the coefficient in Lemma 8 so that its checkpoint term cancels the contribution of db_i from each gap.

Set

$$\boxed{\kappa := \left(\frac{d}{1-d}\right)^\nu = \left(\frac{\epsilon}{c-\epsilon}\right)^\nu.} \tag{16}$$

Theorem 9. *If the scalar hypothesis (14) holds with (16), then for every $n \ge 0$ and $\lambda \ge a$,*

$$\boxed{U_n(\lambda)^\nu \le a^\nu n + \lambda^\nu.} \tag{17}$$

Full proof

We proceed by induction on n. At a maximizing n-step schedule, write $W = U_n(\lambda)$ and choose a nonempty tight set. Applying (11) and Lemma 8 gives

$$W^\nu + \kappa \sum_i [(1-d)b_i]^\nu \le (\lambda + s_{k+1})^\nu + \sum_i (a + s_i + db_i)^\nu.$$

Let m_i be the number of steps in gap i, including m_{k+1} for the final gap. Since k steps were selected, $\sum_i m_i = n - k$. The identities (13) and the induction hypothesis give

Each $m_i < n$, and the parameter $a + db_i$ is at least a, so the induction hypothesis applies.

$$(a + s_i + db_i)^\nu \le a^\nu m_i + (a + db_i)^\nu, \qquad (\lambda + s_{k+1})^\nu \le a^\nu m_{k+1} + \lambda^\nu.$$

Using $(a + db_i)^\nu \le a^\nu + d^\nu b_i^\nu$, we obtain

$$W^\nu + \kappa(1-d)^\nu \sum_i b_i^\nu \le a^\nu n + \lambda^\nu + d^\nu \sum_i b_i^\nu.$$

By (16), $\kappa(1-d)^\nu = d^\nu$. The checkpoint terms cancel exactly, leaving (17).

□

Let $p = 1/\nu$ and take $\lambda = an^p$. Combining (17) with Lemma 4 yields

$$\boxed{\mathcal{R}_n(h) \ge \frac{1}{2^{2p+1}a}\, n^{-p}.} \tag{18}$$

It remains only to determine how close p can be taken to the silver exponent.

Why the Silver Ratio Appears

After moving part of each selected step into its gap and cancelling the resulting terms, it remains to prove a one-variable inequality.

Let $\rho := 1 + \sqrt{2}$, so that $p_{\mathrm{sil}} = \log_2 \rho$. The basic inequality is

$$\boxed{z^{p_{\mathrm{sil}}} + (1-z)^{p_{\mathrm{sil}}} + [z(1-z)]^{p_{\mathrm{sil}}} \le 1, \qquad 0 \le z \le 1.} \tag{19}$$

At the balanced split $z = 1/2$, equality in (19) is

$$\frac{2}{\rho} + \frac{1}{\rho^2} = 1,$$

which is exactly $\rho^2 = 2\rho + 1$. This identity, which also appears in the recursive silver schedule, determines the exponent in our lower bound.

Proof of the silver inequality

Put $\alpha := p_{\mathrm{sil}} - 1$. On $z \in (0, 1/2)$, the derivative of the left side of (19) has the same sign as

$$\phi(z) := (1-z)^{-\alpha} - z^{-\alpha} + 1 - 2z.$$

Moreover,

$$\phi''(z) = \alpha(\alpha+1)\big[(1-z)^{-\alpha-2} - z^{-\alpha-2}\big] < 0.$$

We have $\phi(0+) = -\infty$, $\phi'(0+) = +\infty$, $\phi(1/2) = 0$, and $\phi'(1/2) = \alpha 2^{\alpha+2} - 2 < (1/3)2^{7/3} - 2 < 0$. Hence ϕ' crosses zero once, and ϕ first rises and then falls. Since ϕ is positive just to the left of $1/2$, it crosses zero exactly once. The function in (19) therefore first decreases and then increases on $[0, 1/2]$. Its maximum occurs at 0 or $1/2$, where its value is one. Symmetry completes the proof.

□

Allowing for $\epsilon > 0$

The local hard function requires $\epsilon > 0$, so we need a slightly stronger inequality. For $p \in [p_{\mathrm{sil}}, 3/2]$ and $0 < \chi \le 2$, define $G_{p,\chi}(z) := z^p + (1-z)^p + \chi[z(1-z)]^p$. The silver inequality implies

$$G_{p,\chi}(z) \le 1 - \big[p - p_{\mathrm{sil}} - (\chi - 1)_+\big] z(1-z). \tag{20}$$

Indeed, $-\partial_p G_{p,1}(z) \ge z(1-z)[z^{p-1} + (1-z)^{p-1}] \ge z(1-z)$, while replacing the coefficient 1 by χ increases the expression by at most $(\chi-1)_+ z(1-z)$.

> **Lemma 10.** *Let $p \in [p_{\mathrm{sil}}, 3/2]$, $\nu = 1/p$, $0 < \chi \le 2$, and $\tau := \epsilon^{1/p} \le 1/10$. If*
>
> $$\boxed{p - p_{\mathrm{sil}} \ge (\chi - 1)_+ + 6\epsilon^{1/p},} \tag{21}$$
>
> *then the scalar hypothesis (14) holds for $\kappa = (\epsilon/\chi)^{1/p}$.*

Full proof

Suppose $w = x + y + B - a$ and $wB = \chi xy$. Put $u = x/w$ and $v = y/w$. Since $x, y \ge a$ and $B > 0$, we have $0 < u, v < 1$. Moreover,

$$u + v + \chi uv = 1 + \frac{a}{w} \geq 1.$$

After dividing (14) by w^{ν}, the desired inequality becomes

$$u^{1/p} + v^{1/p} - \tau(uv)^{1/p} \geq 1. \tag{22}$$

Assume otherwise and set $X = u^{1/p}$, $Y = v^{1/p}$. Then $Y < (1 - X)/(1 - \tau X) =: A$. Since $0 \leq A \leq 1$, the mean value theorem gives

$$\left[A^p - (1 - X)^p\right](1 + \chi X^p) \leq \frac{p\tau(1 + \chi)}{1 - \tau} X(1 - X) \leq 6\tau X(1 - X).$$

Consequently,

$$u + v + \chi uv < X^p + A^p + \chi X^p A^p \leq G_{p,\chi}(X) + 6\tau X(1 - X) \leq 1,$$

where the last inequality follows from (20) and (21). This contradicts $u + v + \chi uv \geq 1$.

□

Completing the Proof

Fix $0 < \eta \leq 1/10$, and choose

$$p := p_{\mathrm{sil}} + \eta, \qquad \gamma := \frac{\eta}{4}, \qquad \epsilon := \left(\frac{\eta}{16}\right)^p, \qquad c := \sqrt{1 + 2\gamma + \epsilon^2}, \qquad \chi := c - \epsilon.$$

Then $c \leq 1 + \eta/3$, $\chi \leq 2$, and $\epsilon^{1/p} = \eta/16$. Hence

$$(\chi - 1)_+ + 6\epsilon^{1/p} \leq \frac{\eta}{3} + \frac{3\eta}{8} = \frac{17\eta}{24} < \eta = p - p_{\mathrm{sil}}.$$

Lemma 10 verifies (14). Theorem 9 and (18) then give a polynomial lower bound with exponent $p_{\mathrm{sil}} + \eta$. For $\eta > 1/10$, apply the same conclusion with $\eta' = 1/10$.

The subpolynomial loss

The constant a grows as $\eta \to 0$, and we now bound its growth. In the construction above, take

$$R = \sqrt{\epsilon^2 + (1 + \gamma)^2}, \quad H = \left[\frac{c(1 + \gamma + R)}{\epsilon(\gamma + R)}\right]^{R/\gamma} - 1, \quad C_0 = \epsilon + \epsilon^{-1} + H/\epsilon, \quad B_0 = 2(1 +$$

and choose

$$a := \max\{2, 2C_0/c, 2B_0\}.$$

Under (23), the exponent R/γ is $O(1/\eta)$, while the logarithm of the base in the definition of H is $O(\log(1/\eta))$. Therefore

$$\log a = O\big(\eta^{-1}\log(1/\eta)\big).$$

Substituting this in (18) gives

$$\mathcal{R}_n(h) \geq n^{-p_{\mathrm{sil}}} \exp\big\{-\eta \log n - O\big(\eta^{-1}\log(1/\eta)\big)\big\}.$$

Taking $\eta = \sqrt{\log\log n / \log n}$ proves Theorem 1. Here the hard function and its parameters may depend on the finite horizon n, as allowed in the non-anytime worst case.

Matching the silver upper bound

For $n_k = 2^k - 1$, Altschuler and Parrilo's Theorem 1.1 gives the following bound for the silver schedule, with $\rho = 1 + \sqrt{2}$:

$$\mathcal{R}_{n_k}(h^{\mathrm{sil}}) \leq \frac{1}{1 + \sqrt{4\rho^{2k} - 3}} = O(n_k^{-p_{\mathrm{sil}}}).$$

Wang et al. later determined the exact worst-case objective-gap bound for this schedule, improving the constant without changing the exponent.

For an arbitrary horizon n, take $m = 2^{\lfloor \log_2(n+1) \rfloor} - 1 \leq n$, run the m-step silver schedule, and append $n - m$ zero steps. Since $m = \Theta(n)$, this gives $r_n^* \leq C_2 n^{-p_{\mathrm{sil}}}$. Together with Theorem 1, this shows that $r_n^* = n^{-p_{\mathrm{sil}} + o(1)}$ and $\lim_{n\to\infty} -\log r_n^* / \log n = p_{\mathrm{sil}}$.

The Anytime Lower Bound

We now prove Theorem 2. Fix any $p > p_{\mathrm{sil}}$, put $\nu := 1/p$, and choose a, c, ϵ as in the proof of Theorem 1. Also set $d := \epsilon/c$. We first bound the total stepsize at times when the last step is the largest so far. We follow the record-time approach of Tsai et al. and Jung et al., but the additional ϵ in our cost requires a different counting estimate.

For an infinite schedule $H = (h_t)_{t\geq 1}$, write

$$S_n := \sum_{t=1}^{n} h_t, \qquad M_n := \max_{1\leq t\leq n} h_t, \qquad r_n := \mathcal{R}_n(H_n).$$

We call n a *record time* if $h_n = M_n$.

Lemma 11. *For every fixed $p > p_{\mathrm{sil}}$, there are constants $C_p < \infty$ and $r_{0,p} > 0$ such that every record time n with $M_n \geq B_0$ and $r_n \leq r_{0,p}$ satisfies*

$$\boxed{S_n \leq C_p n M_n^{1-\nu}.}$$

Full proof of Lemma 11

Fix such a record time and abbreviate $M := M_n = h_n$, $R := r_n$, $\xi := H_{n-1}$, and $\Lambda := a + dM$. Let $V := V_\Lambda(\xi)$. The deletion argument in the proof of Lemma 4 shows that a set attaining V selects only steps larger than $a/2 \geq B_0$. Add the final step M to this set. Since

$$c(a+s) + \epsilon M = c(\Lambda + s),$$

Theorem 3 therefore gives

$$R \geq \frac{M^2}{4c^2 V^2}, \qquad V \geq \frac{M}{2c\sqrt{R}}.$$

By decreasing $r_{0,p}$ if necessary, we may therefore assume $V \geq 2dM$.

We next count the entries of ξ above a threshold. For $0 < u \leq M$, let $N_\xi(u) := |\{t < n : h_t > u\}| = k$, list these entries in chronological order as $b_1, \dots, b_k$, and let $\xi^{(0)}, \dots, \xi^{(k)}$ be the intervening blocks, of lengths $m_0, \dots, m_k$. Thus $\sum_{j=0}^{k} (m_j + 1) = n$.

Choose an optimal checkpoint set inside each block and also select $b_1, \dots, b_k$. The cost of this checkpoint set gives

$$V_{a+dM}(\xi) \leq V_{a+dM}(\xi^{(k)}) \prod_{i=1}^{k} \frac{c}{b_i} V_{a+db_i}(\xi^{(i-1)}).$$

Indeed, multiplying the terminal factor $a + s + db_i$ of the preceding block by c/b_i produces $\epsilon + c(a+s)/b_i$, the factor associated with b_i. We use $b_1, \dots, b_k$ only to factor the cost in this way. These steps need not satisfy $b_i \geq B_0$, which is required when applying Theorem 3.

Theorem 9 and $V_\mu \leq U_{m_j}(\mu)$ give

$$\frac{c}{b_i} V_{a+db_i}(\xi^{(i-1)}) \leq \epsilon \left(1 + \frac{a^\nu (m_{i-1} + 1)}{d^\nu b_i^\nu}\right)^{1/\nu},$$

while

$$V_{a+dM}(\xi^{(k)}) \leq dM \left(1 + \frac{a^\nu (m_k + 1)}{d^\nu M^\nu}\right)^{1/\nu}.$$

Here we used $(a+db)^{\nu} \le a^{\nu} + d^{\nu} b^{\nu}$ and $cd = \epsilon$. Since every $b_i > u$, $M \ge u$, and $1 + z \le e^{z}$, it follows that

$$\frac{V}{dM} \le \epsilon^{k} \exp\!\left\{ \frac{a^{\nu}}{\nu d^{\nu}}\, n u^{-\nu} \right\}.$$

Combining this with $V/(dM) \ge 2$ and taking logarithms yields

$$N_{\xi}(u) \le \frac{a^{\nu}}{\nu d^{\nu} \log(1/\epsilon)}\, n u^{-\nu}.$$

Integrating the counting function and using $0 < \nu < 1$ gives

$$\sum_{t=1}^{n-1} h_t = \int_{0}^{M} N_{\xi}(u)\, du \le C_p n M^{1-\nu}.$$

It remains to include the final step M. Combining the lower bound on V with $V \le U_{n-1}(a + dM)$ gives

$$(2c\sqrt{R})^{-\nu} M^{\nu} \le a^{\nu}(n-1) + (a+dM)^{\nu} \le a^{\nu} n + d^{\nu} M^{\nu}.$$

For sufficiently small $r_{0,p}$, the last term is absorbed into the left-hand side, and hence $M^{\nu} \le C_p n$. Therefore $M \le C_p n M^{1-\nu}$, which completes the proof.

□

Completing the proof of Theorem 2

First fix $p > p_{\mathrm{sil}}$, put $\delta := p - p_{\mathrm{sil}}$, and set

$$\beta(p) := \frac{2p}{1+p} = \frac{2}{1+\nu}.$$

Selecting no checkpoints in Theorem 3 gives $r_n \ge 1/[4(1+S_n)]$. If the infinite schedule is bounded, then $S_n = O(n)$, which is already stronger than the conclusion of Theorem 2. We therefore assume that the schedule is unbounded and consider its arbitrarily large strict record times.

At a record time to which Lemma 11 applies, selecting only the final step in Theorem 3 gives

$$r_n \ge \frac{1}{4}\left[\frac{M_n}{c(a + S_{n-1}) + \epsilon M_n} \right]^{2}.$$

For small r_n, the term containing ϵM_n can be moved to the left. Since $S_n \ge M_n \ge B_0$, we have $a + S_{n-1} \le (1 + a/B_0) S_n$, and therefore

$$M_n \le C_p S_n \sqrt{r_n}.$$

Selecting no checkpoints also gives $S_n \ge 1/(8 r_n)$ for all sufficiently large record times.

Substituting this into Lemma 11 gives

$$S_n \le C_p n (S_n \sqrt{r_n})^{1-\nu}, \qquad S_n^{\nu} \le C_p n r_n^{(1-\nu)/2}.$$

Using $S_n \ge 1/(8r_n)$ once more yields

$$1 \le C_p n r_n^{(1+\nu)/2}.$$

Hence

$$r_n \ge c_p n^{-\beta(p)}.$$

We now track the dependence on δ. The parameter estimates in the proof of Theorem 1 and the proof of Lemma 11 give

$$-\log c_p = O\big(\delta^{-1}\log(1/\delta)\big), \qquad \log(1/r_{0,p}) = O\big(\delta^{-1}\log(1/\delta)\big), \qquad \log B_0 = O(\log(1/\delta)),$$

Moreover, $\beta(p) = p_{\mathrm{any}} + O(\delta)$.

At each sufficiently large strict record time n, take

$$\delta = \sqrt{\frac{\log\log n}{\log n}}, \qquad p = p_{\mathrm{sil}} + \delta.$$

These parameters may vary with n, since $\mathcal{R}_n(H_n)$ takes a separate supremum over hard functions at each horizon. If $M_n < B_0(\delta)$, then $S_n \le n B_0(\delta)$, and the bound obtained by selecting no checkpoints is stronger than the desired estimate. The same is true if $r_n > r_{0,p}$. In every other case, the preceding argument gives

$$r_n \ge n^{-p_{\mathrm{any}}} \exp\big\{-C\left(\delta\log n + \delta^{-1}\log(1/\delta)\right)\big\}.$$

With the displayed choice of δ, the exponent on the right is $O(\sqrt{\log n\ \log\log n})$. Since an unbounded schedule has infinitely many strict record times, this proves Theorem 2.

Concluding Remarks

Our proof builds on the checkpoint construction of Jung et al. and reduces the local transfer coefficient to a value arbitrarily close to one. We handle the additional term ϵb_i by moving part of each checkpoint into the preceding gap. The weighted splitting lemma introduces a term that cancels the resulting contribution from these gaps. The silver inequality (19) then determines the convergence exponent.

The silver rate is therefore tight at the level of polynomial exponents. The proof does not establish $\Omega(n^{-p_{\mathrm{sil}}})$. At $p = p_{\mathrm{sil}}$, the balanced split in (19) is an equality, leaving no fixed slack for $\epsilon > 0$. Removing the subpolynomial loss remains open.

Theorem 2 likewise determines the optimal anytime polynomial exponent, but it does not prove the endpoint statement $\limsup_{n\to\infty} n^{p_{\mathrm{any}}}\mathcal{R}_n(H_n) > 0$.

Bibliographic Information

```
@misc{ye2026silver,
  author = {Yuhan Ye and Kaizhao Liu},
  title  = {The Silver Rate Is (Almost) Tight},
  year   = {2026},
  note   = {Blog post},
  url    = {https://yeyuhanyyh.github.io/gd-silver-rate/}
}
```

For the full paper, cite the arXiv preprint:

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@misc{ye2026silverrate,
  author        = {Yuhan Ye and Kaizhao Liu},
  title         = {Silver Rate Is (Almost) Optimal for Gradient Descent},
  year          = {2026},
  eprint        = {2609.09152},
  archivePrefix = {arXiv},
  primaryClass  = {math.OC},
  doi           = {10.48550/arXiv.2609.09152},
  url           = {https://arxiv.org/abs/2609.09152}
}
```

AI Disclosure. We carefully read Jung et al.'s paper and traced the difference between their exponent $\log_2(1+\sqrt{3})$ and the silver exponent $\log_2(1+\sqrt{2})$ to the factor 2 in their local transfer estimate. Their local Huber component has a fixed gradient direction. It changes both adjacent coordinates, so both changes reduce the margin that keeps the component active. We wondered whether we could reduce this loss by bending the local trajectory, gradually turning the descent direction toward x_{i+1} as the remaining amplitude passes from x_i to x_{i+1}. We shared this intuition with ChatGPT-6 Astra. Through several rounds of substantive interaction and detailed calculations by ChatGPT-6 Astra Ultra, we designed a smooth convex hard function whose local gradient rotates along a circular arc and whose gap coefficient approaches 1. Combining this construction with a refinement of Jung et al.'s recursive analysis gave the present result. The paper is available as an arXiv preprint.

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